

Miaohui Shi

施妙辉

Algorithm Engineer | High-performance Computing & Robotics

Taizhou, Zhejiang, China | [shimiaohui0426.github.io](https://github.com/shimiaohui0426)

[GitHub](#) / [ShiMiaohui0426](#) • [LinkedIn](#) / [Miaohui Shi](#)

Algorithm engineer working across high-performance computing, computational lithography and robotics. Develop C++ / Python systems for GDS rasterization, multi-robot interaction and vision-based tactile sensing.

800 → 24 min

GDS end-to-end runtime

33.3× incl. platform changes

92 → 44 min

Same-condition server run

2.09× with partitioned reads

601.25 Hz

Tactile force acquisition

Reported sensor-system rate

PROFESSIONAL EXPERIENCE

Taizhou Optoelectronics Industry Innovation Center

Sep 2024 - present

Algorithm Engineer

- Ported GDS rasterization from Python to C++; optimized computation, parallelism, memory access and compilation. Added multi-node processing, CD bias and correction support.
- Migrated a small-PC workload to multi-node servers: from an initial estimate of tens of thousands of hours to a measured 24-minute six-node run. Measured 800 to 24 minutes: 33.3× end-to-end acceleration, including hardware and deployment changes.
- Partitioned reads so each node loads only assigned data: 92 to 44 minutes (2.09×) with the same server configuration and node count. Minute-level comparisons use the same GDS input and 4-pass workload.
- Built software for local inspection of very large raster images and equipment control.

Hesai Technology / Sharp-related R&D

Apr - Aug 2024

Algorithm Development

- At Hesai Technology, developed LiDAR functional-degradation algorithms and trained noise-segmentation models; also contributed to Sharp-related tactile-sensor R&D.
- Designed and trained vision-based tactile-sensor mechanical models; developed software on Ubuntu.

EDUCATION

Waseda University

M.S., Mechanical Engineering

2022-Mar 2024 • Japan

Huazhong University of Science and Technology

B.S., Mechanical Science & Engineering

2017-2021 • China

TECHNICAL SKILLS

Performance & engineering

C++ • Python • Linux / Ubuntu • parallel and distributed processing • memory and I/O optimization • GDS rasterization

Robotics & interaction

ROS • Unity • multi-robot systems • HRI experiment design • PASAT / NASA-TLX • behavioral data analysis

Vision & learning

YOLO • semantic segmentation • anomaly detection • model training and optimization • vision-based tactile sensing

Software & interfaces

C# • Qt • front-end UI development • industrial vision software • TwinCAT / PLC integration

SELECTED RESEARCH

Multi-robot interaction, attention and automation | CASE 2024

Independently developed the experimental system (excluding the mechanical-head module), designed and conducted the experiments, and completed all data analysis. Adapted the PASAT auditory-addition task to a dual-task study involving three robotic arms, with NASA-TLX and user-experience measures.

High-speed vision-based tactile sensing | ICAT-EGVE 2024

Independently developed algorithmic principles, implementation, performance tuning and the front-end UI. The method combines local sparse marker tracking with force estimation using an asymmetric stiffness-coefficient matrix. The paper reports 601.25 Hz sensor-system force acquisition; this is not a standalone algorithm benchmark.

PUBLICATIONS

2024 High-Speed Vision-Based Haptic Sensor for Robotic Dermatological Palpation: Force Sensing Method Using Asymmetric Stiffness Coefficient Matrix

Fumihiro Kato, Miaohui Shi, Kaito Kamishima, Hiroyasu Iwata
ICAT-EGVE 2024

2024 The Role of automation and User Experience in Multi Robot Systems

Kouta Suzuki, Miaohui Shi, Vimolmongkolporn Vitvasin, Yukiko Iwasaki, Hiroyasu Iwata
IEEE CASE 2024 • pp. 3818-3825

2023 Balancing Real-world Interaction and VR Immersion with AI Vision Robotic Arm

Xiaoran Yang, Yang Zhan, Yukiko Iwasaki, Miaohui Shi, Shijie Tang, Hiroyasu Iwata
IEEE ICMA 2023 • pp. 2051-2057

2021 基于TwinCAT的机器人实验平台设计

黄弢, 施妙辉, 汪迪, 王峻峰
实验技术与管理 • 38(12):197-202+207

2021 A Remote Experiment Platform based on HTML5 for Industrial Robots

Tao Huang, Miaohui Shi
MED 2021 • pp. 920-923

PATENTS

Eight distinct patent application records are listed on the website, covering visual inspection, material handling, subpixel image processing and vision-based tactile sensing. Publication and grant versions of the same application are counted once.

Full patent record / 8 applications

EARLIER ACADEMIC EXPERIENCE

Oct-Dec 2022 | Learning Assistant • Waseda University

Dec 2019-Feb 2022 | Research Assistant • Huazhong University of Science and Technology

RESEARCH DIRECTION

Future doctoral research interest: household service robots, tactile perception, human-robot collaboration and appropriate automation. Industry interests include EDA, computational lithography and high-performance algorithm engineering.